

Statistical-mechanical characterization of collective dynamics in decentralized LLM-agent networks

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Abstract

We construct a finite interacting system whose microscopic variables are realized choices made by independent large-language-model agents. Agent i owns a categorical belief $b_i \in \{-1, +1\}$, an executed action $a_i \in \{-1, +1\}$, confidence, commitment, bounded memory, a private field, workload, inbox, outbox, and typed local authority. Random-sequential updates are generated by a pinned instruction model from only local state and delivered messages. We connect a measured single-agent response law to collective order, susceptibility, correlations, entropy rate, multi-information, an effective symmetric-layer energy, relaxation, and time-reversal divergence. A prospective evaluation uses 32,672 decisions in 72 graph trajectories, followed by fixed field, communication-partition, and message-corruption quenches. Increasing coupling changes absolute order by -0.0006 (95% interval -0.0069 to 0.0057); increasing decoding noise changes it by 0.0078 (0.0013 to 0.0152). These coupling and noise contrasts do not replicate the V12 discovery directions. In contrast, bounded memory increases adjusted block irreversibility by 0.04030 nats per attempted update (0.02883 to 0.05856), replicating the discovery direction. A fixed multivariate macrostate strongly resolves field reversal but only partly separates communication partition, message corruption, and nominal operation; its four-class accuracy is 0.500. These are finite-size, single-model results. Reference energy is an effective observable, decoding noise is not physical temperature, and reversal divergence is a coarse-grained irreversibility measure rather than exact thermodynamic entropy production.

Keywords: statistical mechanics of agents; kinetic Ising models; large language models; entropy rate; irreversibility; quench dynamics.

1 Introduction

Networks of autonomous language-model agents are emerging interacting many-body systems. Each component uses a complicated language-conditioned stochastic rule, retains private state, receives only selected messages, and acts with bounded local authority. The microscopic rule is not a spin-flip formula, yet the network produces familiar collective phenomena: alignment, fragmentation, fluctuations, persistence, and recovery after changes in fields or links. A central question for statistical physics is therefore not whether a thermodynamic metaphor can be attached after simulation, but whether a compact set of well-defined observables links local response to collective trajectories and makes distinct regimes quantitatively interpretable.

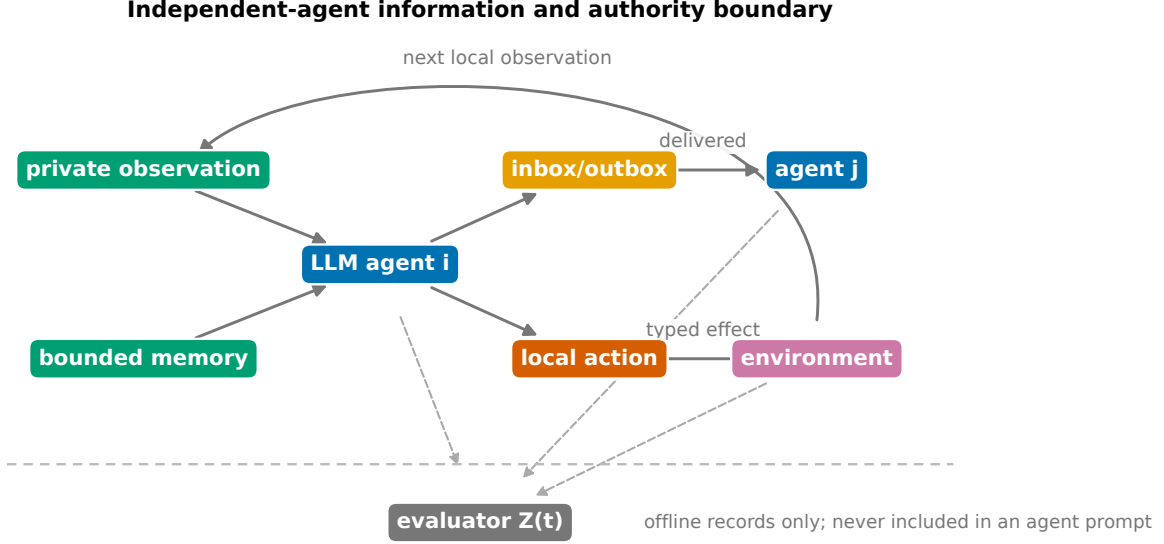


Figure 1: Independent-agent information and authority boundary. The scheduler offers an update and transports the model-selected packet; it cannot inspect peer-private state or substitute a belief, action, commitment, memory, signal, or tool. The reduced state Z_t is evaluator-only.

The kinetic Ising model demonstrates how stochastic local updates generate order and relaxation [1]. Logit dynamics has a related equilibrium interpretation when strategic interactions derive from a potential [2]. Away from detailed balance, state-space currents and path reversal distinguish stationary irreversibility [3–5]. These exact constructions provide useful references, but an LLM network violates several assumptions: the visible binary state may not close under the dynamics; natural-language messages can carry hidden semantic variables; memory induces history dependence; and decoding temperature controls token sampling rather than a physical bath.

Recent studies have nevertheless found compact collective structure in LLM populations. Network topology and social weighting alter conformity [6]; intrinsic field bias can dominate effective cooperation [7]; noise and conformity produce finite-size debate crossovers [8]; population size changes basins of attraction [9]; and large collections of communities exhibit a small number of dynamical regimes predicted by a fitted statistical model [10]. The present study is complementary. It preserves explicit private-state and message boundaries, separates belief from executed action, measures trajectory-level information and fluctuation observables, prospectively tests parameter effects from an earlier discovery study, and applies fixed quenches to the same interacting agents.

Our objective is characterization, not performance optimization. We ask how a measured neighbor response aggregates into order and fluctuations; whether coupling and decoding noise reproduce discovery directions; whether bounded memory adds time asymmetry; how field, topology, and channel disruptions trace paths through a common reduced state; and which information is missed by state counts or magnetization alone. No observable is required to improve task reward, and no human or field-system inference is made.

2 Statistical mechanics of agent collectives

Statistical mechanics is useful here at three distinct levels. First, realized binary choices define a discrete configuration on a graph. Second, empirical response functions can be approximated by kinetic models and analyzed by mean-field or stability methods. Third, order, fluctuations, entropy, energy, correlations, and time asymmetry reduce a high-dimensional trajectory to a small set of interpretable coordinates. Only the first two ingredients are needed for a rigorous finite stochastic process; no assertion that an LLM is a physical thermodynamic object follows.

Information quantities retain their literal mathematical meanings. Shannon entropy measures uncertainty of a specified distribution [11]. Total correlation measures dependence not captured by marginal entropies [12]. Transfer entropy measures directed predictive information [13]; it is not entropy production. For a stationary Markov chain with observed transition kernel, probability-current entropy production has a precise definition. When visible variables omit history, forward/reverse path divergence is instead a coarse-grained lower-bound diagnostic [14, 15]. Finite-sample entropy estimators are biased, particularly under sparse support, so all such quantities require fixed coarse-graining and null floors [16].

2.1 Relation to kinetic, nonreciprocal, and quench models

The closest analytical reference is an asymmetric kinetic Ising system, for which mean-field closures can be organized around response functions and delayed correlations [17]. Our measured transition rule differs in two ways. Its coefficients are induced by a language-conditioned model rather than specified as a spin kernel, and belief and action form distinct but coupled layers. We therefore test a fitted kinetic closure against direct trajectories instead of assuming it by construction.

Nonreciprocal couplings can generate phenomena absent from equilibrium Hamiltonian systems [18], and their entropy production can be defined for suitable stochastic dynamics [19]. The immutable V12 study used this theory as a reference but found no reliable incremental irreversibility from degree- and traffic-matched directionality in the actual LLM trajectories. V13 preserves that boundary rather than searching new directed graphs. Its nonequilibrium perturbations are instead controlled quenches in fields, active delivery support, and packet content. These are finite classical stochastic trajectories; quantum-quench results [20] supply terminology about relaxation but not a claimed mapping of microscopic dynamics.

3 Independent LLM-agent microdynamics

3.1 Observable state and autonomy

The local state is

$$x_i(t) = (b_i(t), a_i(t), c_i(t), q_i(t), m_i(t), w_i(t)), \quad (1)$$

where $b_i, a_i \in \{-1, +1\}$ are categorical belief and action, c_i is bounded confidence, q_i a categorical commitment, m_i bounded memory, and w_i local workload. Each persistent identity separately owns these variables, a private field h_i , an inbox, outbox, role, and typed authority. The global record $X_t = (x_1, \dots, x_N)$ is available only to the evaluator.

At each attempted update a random permutation schedule offers one agent a turn. Its prompt is rebuilt from its private observation, local state, explicitly delivered packets, and

bounded memory when enabled. The model chooses belief, action, confidence, commitment, memory state, outgoing categorical signal, and tool. One syntax-repair attempt is permitted; a persistently invalid response leaves the state unchanged rather than invoking a centralized scientific choice. One sweep equals N attempted updates.

The model is Qwen2.5-7B-Instruct at the pinned revision reported in the data package [21]. The primary regime carries no hidden chat history: the prompt is fully reconstructed from recorded current state. A secondary regime adds exactly three local memory summaries. Neither construction proves Markov closure, which is tested empirically.

3.2 Latent-label counterbalancing

Displayed alternatives “amber” and “cobalt” are semantically symmetric. Their order and mapping to latent ± 1 spins are counterbalanced, private fields and disordered starts are balanced, and all responses are decoded to the latent coordinate before analysis. Matched factor and quench arms share graph, initial state, update schedule, display mapping, and inference seeds. Thus a coupling or disruption contrast does not change option identity or message opportunity.

4 Individual response law

The microscopic grid varies private field h_i , prior belief and action, delivered neighbor field, coupling J , and decoding-noise setting τ . The primary measured responses are realized categorical choices; reported confidence is secondary and is not interpreted as a calibrated probability. We fit

$$\Pr[b_i(t+1) = +1 \mid X_t] = \text{logit}^{-1}(\beta_0 + \beta_h h_i + \beta_n J \bar{b}_{\partial i} + \beta_b b_i + \beta_a a_i), \quad (2)$$

$$\Pr[a_i(t+1) = +1 \mid X_t] = \text{logit}^{-1}(\gamma_0 + \gamma_b b_i(t+1) + \gamma_a a_i(t)). \quad (3)$$

The model is descriptive. It supplies susceptibilities and a surrogate, not a normative Bayesian agent or a claim that token temperature equals an effective thermodynamic temperature.

5 Effective kinetic and mean-field models

For a homogeneous approximation, let m_b and m_a be layer means. Replacing the local neighbor field by m_b yields

$$m_b(t+1) \simeq 2f_b(h + Jm_b(t), m_b(t), m_a(t)) - 1, \quad (4)$$

$$m_a(t+1) \simeq 2f_a(m_b(t+1), m_a(t)) - 1. \quad (5)$$

Linearization of this two-dimensional map gives a local stability matrix. The belief component contains the fitted slope multiplied by J , while persistence and belief-action coupling modify both eigenvalues. We report the local slope as a crossover diagnostic, not a critical exponent.

Writing the fitted logits as

$$u_b = \beta_0 + \beta_h h + \beta_n J m_b + \beta_b m_b + \beta_a m_a, \quad (6)$$

$$u_a = \gamma_0 + \gamma_b \tilde{m}_b + \gamma_a m_a, \quad (7)$$

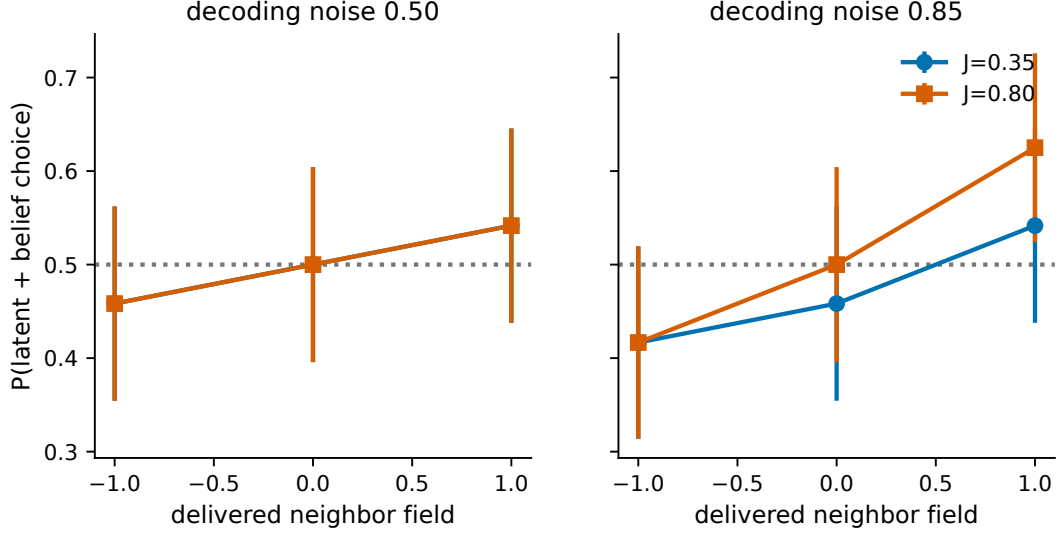


Figure 2: Individual categorical response to delivered neighbor field after latent-label counterbalancing. Points aggregate independently seeded information states; error bars show sampling uncertainty within the frozen microscopic grid.

and $s(u) = 2(1 + e^{-u})^{-1} - 1$, the homogeneous map is $\tilde{m}_b = s(u_b)$ and $\tilde{m}_a = s(u_a)$. Since $s'(u) = 2p(u)[1 - p(u)]$, its local Jacobian is

$$\mathbf{J} = \begin{pmatrix} s'(u_b)(\beta_n \mathbf{J} + \beta_b) & s'(u_b)\beta_a \\ s'(u_a)\gamma_b s'(u_b)(\beta_n \mathbf{J} + \beta_b) & s'(u_a)[\gamma_b s'(u_b)\beta_a + \gamma_a] \end{pmatrix}. \quad (8)$$

The spectral radius $\rho(\mathbf{J})$ is a local persistence diagnostic. A value near one predicts slow response of the fitted homogeneous surrogate; it does not prove singular behavior in either the finite surrogate or the LLM system.

The CPU surrogate applies the fitted stochastic response to independent agents on ring and fixed-degree modular graphs for $N = 8, 16, 32, 64$. It densely maps coupling and decision-noise settings with 16 graph/seed realizations per cell. Direct LLM observations remain the empirical anchors: the surrogate can explain trend directions or miss them, but cannot replace those trajectories.

6 Macroscopic observables

Belief and action order are

$$m_b = N^{-1} \sum_i b_i, \quad m_a = N^{-1} \sum_i a_i, \quad (9)$$

and cross-layer overlap is $q_{ba} = N^{-1} \sum_i b_i a_i$. Disagreement is the fraction of undirected support edges joining unlike spins. Finite-size belief susceptibility is $\chi_b = N \text{Var}(m_b)$, without a temperature normalization. Integrated autocorrelation time is calculated from the positive sequence of the magnetization autocorrelation.

The fixed symmetric-layer reference energy is

$$H_{\text{ref}} = -\frac{J_b}{2} \mathbf{b}^T W_s \mathbf{b} - \frac{J_a}{2} \mathbf{a}^T D_s \mathbf{a} - K \mathbf{a}^T \mathbf{b} - \mathbf{h}^T \mathbf{b} - \mathbf{g}^T \mathbf{a}, \quad (10)$$

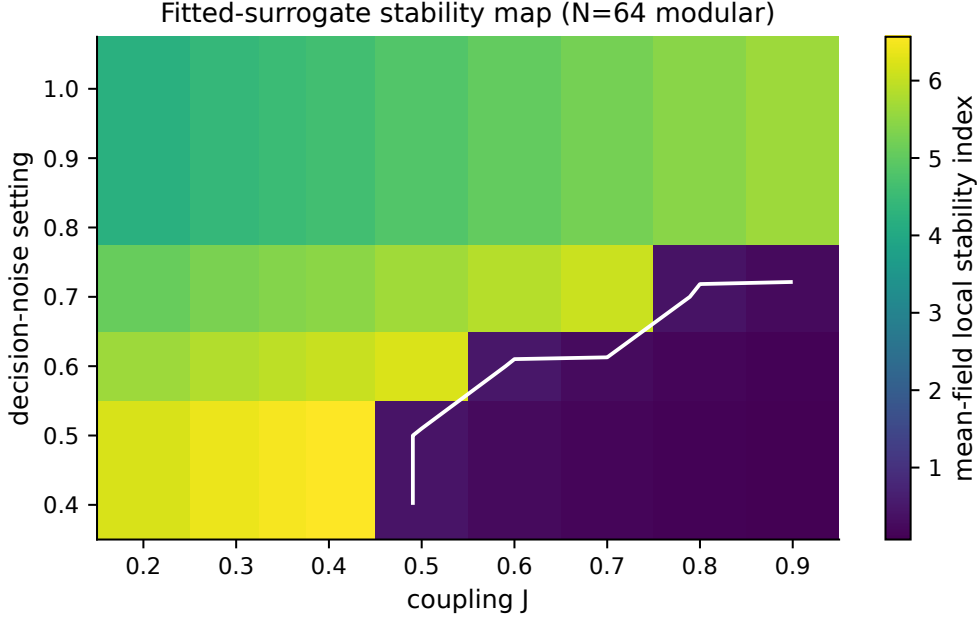


Figure 3: Dense fitted-surrogate stability map. The contour is a local mean-field diagnostic; it does not establish a thermodynamic phase boundary.

with the reported coordinate $e_{\text{ref}} = H_{\text{ref}}/N$. The communication partition changes active delivery but not this reference layer. We call $N \text{Var}(e_{\text{ref}})$ an energy-fluctuation observable, not physical heat capacity.

For large networks, a fixed macrostate binning of $(m_b, m_a, q_{ba}, e_{\text{ref}}, D)$ defines configuration entropy and a first-history conditional entropy rate. Local-word entropy uses each node’s belief, action, and neighbor majority. Total correlation is

$$\mathcal{T} = \sum_i H(X_i) - H(X_1, \dots, X_N), \quad (11)$$

estimated over a five-sweep rolling window. These quantities answer different questions: marginal disorder may be high while dependence is also high, and an entropy increase may represent adaptation, fragmentation, or transient uncertainty depending on order and energy.

The evaluator’s reduced coordinate is

$$Z_t = (m_b, m_a, q_{ba}, S_{\text{config}}, h_{\text{rate}}, \mathcal{T}, e_{\text{ref}}, v_E, \chi_b, \rho, D). \quad (12)$$

It is a prospectively fixed observable representation, not a complete thermodynamic state.

7 Prospective design and inference

V12 is the discovery study and remains immutable. Its coupling contrast increased absolute belief magnetization by 0.03298, susceptibility by 0.04550, and integrated correlation time by 2.319 attempted updates. Its decoding-noise contrast reduced these quantities by 0.05387, 0.07356, and 5.990 updates. Bounded memory increased adjusted block irreversibility by 0.01790 nats per update. Conversely, traffic- and degree-matched nonreciprocity did not reliably increase irreversibility. V13 was designed to test the coupling, noise, and memory directions independently, without reopening the nonreciprocity claim.

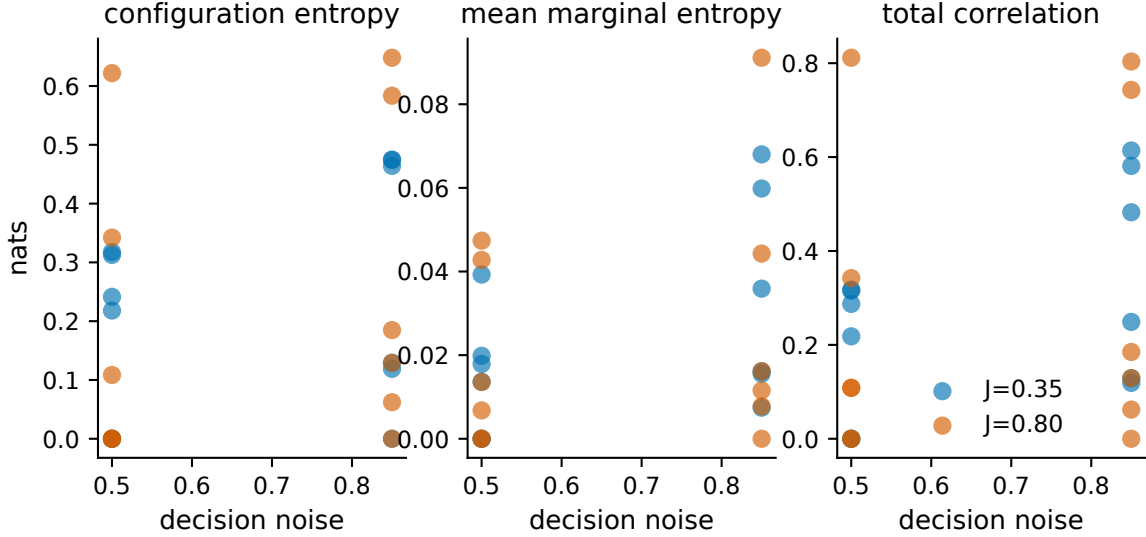


Figure 4: Configuration entropy, mean single-variable marginal entropy, and total correlation separate configurational diversity from collective dependence.

Work Package A crosses $N \in \{8, 16\}$, $J \in \{0.35, 0.80\}$, and $\tau \in \{0.50, 0.85\}$ on modular graphs, with a ring subset and ordered-start relaxation subset. Work Package B pairs Markovized and bounded-memory $N = 16$ agents through a 14/14/14-sweep field reversal. Work Package C pairs an undisturbed control with field reversal, removal of inter-community delivery, and corruption of exactly half of categorical packets; each uses 15 sweeps of baseline, disruption, and recovery.

The inferential unit is the matched graph/environment trajectory cluster. Updates, agents, messages, windows, and tokens are never independent replicates. H1 and H2 use intersection–union directional tests over order, susceptibility, and correlation time. H3 tests the paired memory contrast. These three hypothesis-level p values receive Holm correction. Confidence intervals use 10,000 graph-cluster bootstrap replicates. H4–H7 are separately predeclared new V13 analyses.

8 Collective order, fluctuations, and relaxation

H1 was not supported under its frozen criterion. For $J = 0.80$ minus 0.35, absolute order changed by -0.0006 (95% interval -0.0069 to 0.0057), susceptibility by -0.00345 (-0.01340 to 0.00878), and integrated correlation time by -3.556 attempted updates (-8.910 to -0.237). H2 was not supported under its frozen criterion. For $\tau = 0.85$ minus 0.50, the corresponding changes were 0.0078 (0.0013 to 0.0152), 0.01595 (0.00252 to 0.03310), and 3.232 updates (0.752 to 5.484). Thus both confirmatory contrasts point opposite to their V12 discovery directions for the outcomes with intervals excluding zero; their Holm-adjusted hypothesis-level p values are 1.0. Graph-level points, ring replication, and ordered-start relaxation are retained in the result tables.

The direct sizes are insufficient for a Binder-crossing or scaling-collapse claim. We therefore use “collective-regime change” or “finite-size crossover,” not phase transition. The dense surrogate map extends size only as an explanatory extrapolation and is visually distinguished from direct LLM anchors.

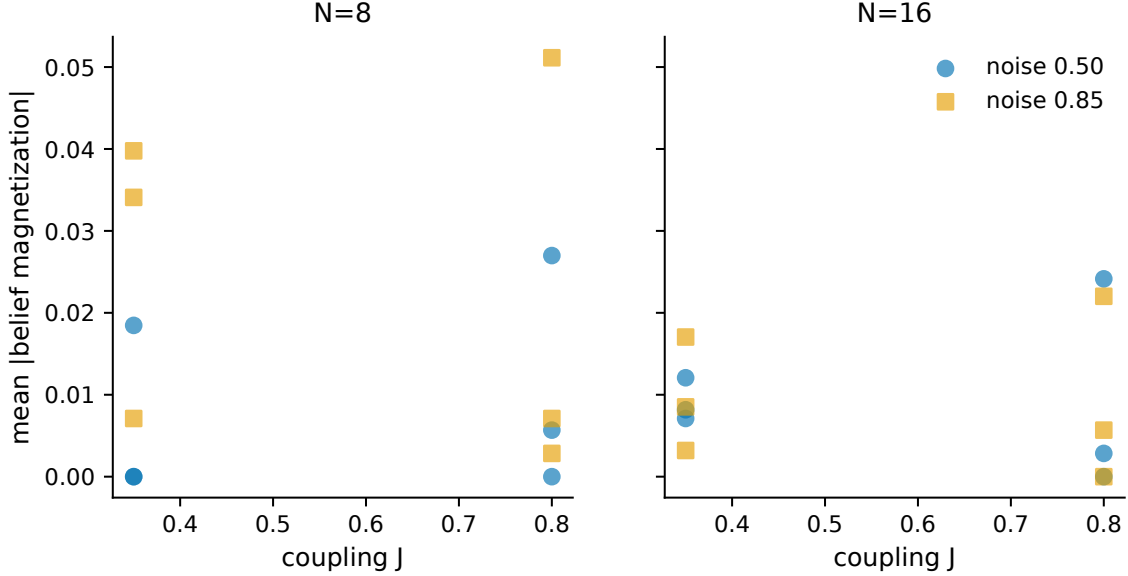


Figure 5: Direct LLM finite-size order across the prospectively fixed coupling–noise anchors. Every point is a complete graph trajectory; no time step is treated as an independent replicate.

9 Entropy and effective-energy representation

Order and energy are correlated but not interchangeable. Configurations with similar magnetization can differ in domain arrangement, action alignment, multi-information, and temporal predictability. Likewise, a shift in reference energy can arise from field mismatch, action–belief inconsistency, or spatial disagreement. Joint energy–entropy trajectories therefore expose degrees of freedom hidden by state counts alone.

We omit a scalar free-energy coordinate. The microscopic response provides a decision-noise control and fitted slopes, but no unique, reproducible physical temperature mapping. The two-dimensional energy–entropy plane is therefore more transparent and avoids implying a Lyapunov function.

10 Persistent memory and irreversibility

For the frozen macrostate word Y_t , the length- L reversal statistic is

$$\mathcal{I}_L = \frac{1}{L-1} D_{\text{KL}}[p(Y_t, \dots, Y_{t+L-1}) \| p(Y_{t+L-1}, \dots, Y_t)]. \quad (13)$$

We subtract the mean from 200 time-shuffled paths of identical length and state occupancy. The result is reported in nats per attempted update. We also measure history-order conditional mutual information. Nonzero history dependence means that $\mathcal{I}_L$ remains a coarse-grained irreversibility diagnostic, not an exact total entropy-production rate.

H3 was supported under its frozen criterion. The paired memory contrast was 0.04030 nats/update (95% interval 0.02883 to 0.05856). Field, graph, schedule, mapping, initial state, and inference seeds are shared within each pair. Memory effects on persistence, loop area, and post-quench recovery are secondary, avoiding the inference that all time asymmetry is beneficial or dissipative.

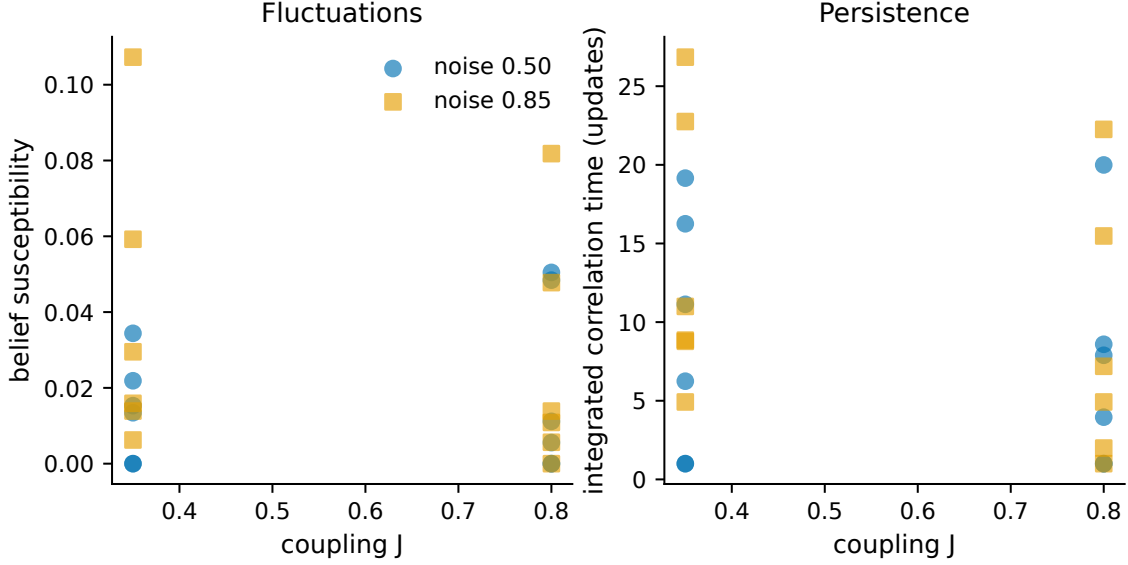


Figure 6: Belief susceptibility and integrated correlation time in the modular primary design. These are finite-size response and persistence observables.

11 Controlled quenches and recovery

The three disruptions isolate environmental, topological, and informational changes. Field reversal changes every private field and then restores it. Partition removes cross-community delivery paths while preserving within-group communication and message opportunities. Corruption flips categorical content for a preassigned half of senders while preserving payload length and delivery count. Agents see only the resulting local evidence; the disruption label is not prompted.

A leave-one-cluster-out nominal manifold is fit from undisturbed trajectories using a ridge-regularized Mahalanobis metric on Z_t . For abrupt quenches we measure response—not early warning: maximum departure, path length, time outside the nominal region, overshoot, recovery time, and final residual distance. H4 was supported under its frozen criterion. The average disruption-minus-control maximum departure was 14.545 (95% interval 14.055 to 14.867). This aggregate is dominated by field reversal: mean maximum distance was 45.055 for reversal, compared with 2.315 for message corruption, 1.633 for partition, and 1.789 for the nominal control. Accordingly, H4 establishes a response to the environmental-field quench, not a generic response to every disruption class.

12 Quantifying information in the reduced state

Three representations are frozen before formal execution. The simple model uses mean belief, mean action, confidence, and message count. The order-only model uses magnetizations, overlap, and disagreement. The full model adds configuration entropy, entropy rate, total correlation, effective energy, energy fluctuations, susceptibility, spatial correlation, and nominal-manifold distance. A regularized multinomial logistic model is trained under leave-one-graph-cluster-out cross-validation; no observable is selected on a held-out trajectory.

H5 was supported under its frozen criterion. Full-representation four-class accuracy was 0.500. H6 was supported under its frozen criterion. Its paired increment over the strongest reduced representation was 0.188 (95% interval 0.062 to 0.250). All four field-

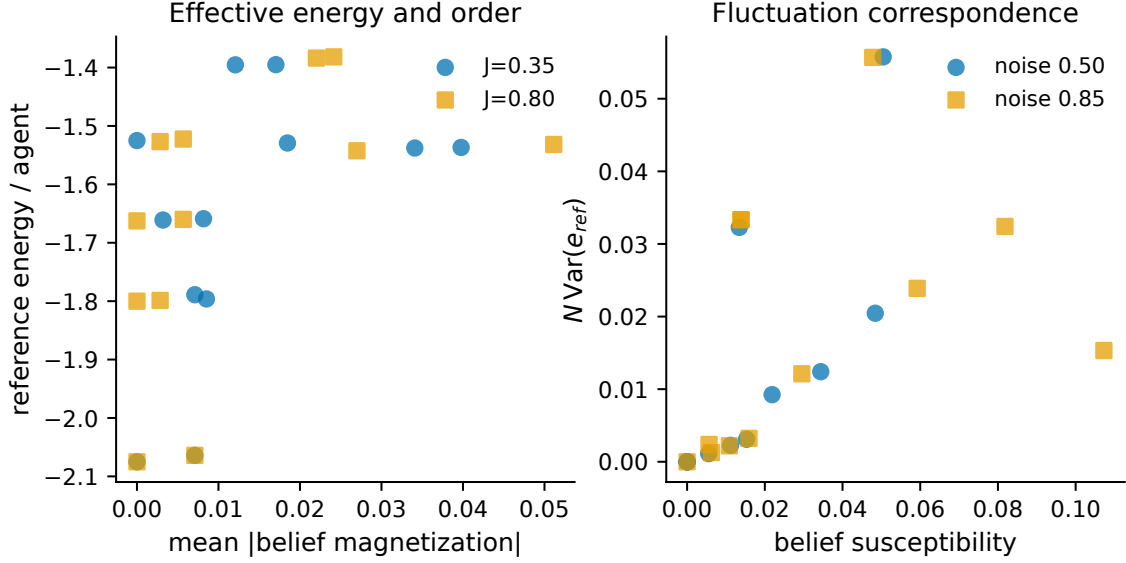


Figure 7: Effective reference energy versus order and its fluctuation analogue versus susceptibility. The energy is defined from the symmetric empirical influence layer and is not literal physical energy.

reversal panels were classified correctly, whereas only four of the twelve partition, corruption, and nominal panels were. With four held-out clusters, these frozen criteria are therefore positive but preliminary: they demonstrate partial separation rather than comprehensive disruption identification. This is not a claim that entropy outperforms a KPI controller. It asks whether jointly measured collective fluctuation, dependence, energy, and temporal coordinates expose trajectory structure absent from averages.

13 Effective-model comparison

H7 was not supported under its frozen criterion. The surrogate coupling contrast was $+0.01497$, whereas the direct LLM contrast was -0.00059 ; the surrogate therefore missed the coupling direction. It captured the positive noise direction but overestimated its magnitude ($+0.02955$ versus $+0.00775$). The effective kinetic model is thus a partial descriptive approximation rather than an explanation of both primary factor trends. Semantic asymmetry, finite microscopic sampling, memory, option-order effects, and action–belief coupling remain candidate residual coordinates.

14 Limits of thermodynamic interpretation

Several distinctions are essential. First, binary coordinates are measured choices, but the LLM’s hidden computation is not a binary spin. Second, H_{ref} is an effective observable defined by the analyst; no Gibbs stationary distribution is asserted. Third, decoding temperature alters token sampling and is called decision noise. Fourth, total correlation and entropy rate are information measures whose estimates depend on finite windows and coarse-graining. Fifth, time-reversal divergence becomes exact Markov entropy production only under closure and stationarity assumptions not generally met by these data. Finally, $N = 8, 16$ direct networks cannot establish a thermodynamic limit.

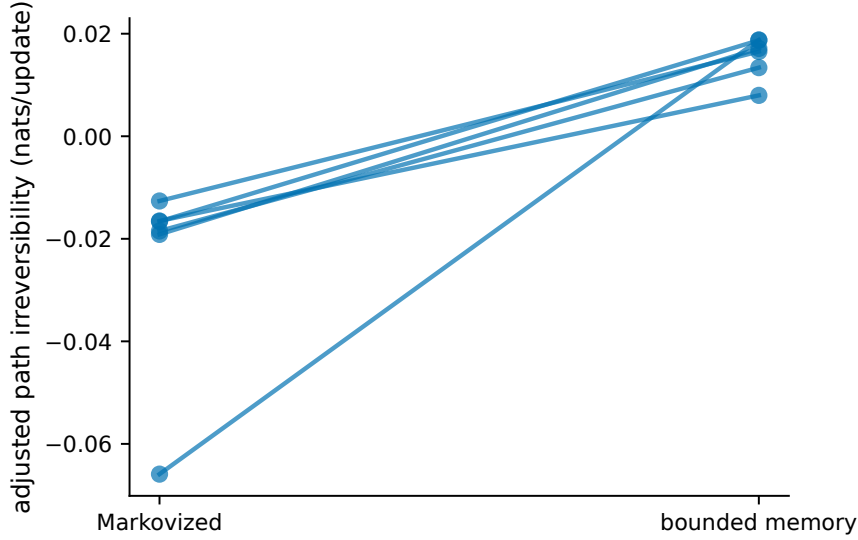


Figure 8: Matched Markovized and bounded-memory pathwise irreversibility. Lines join the same graph/environment cluster.

The fixed-degree modular topology has a perfect cross-community matching. This is analytically convenient and makes partition restoration exact, but is not a generic social graph. One open-weight model, symmetric abstract alternatives, and short bounded outputs limit external validity. Natural-language semantics are deliberately constrained to obtain a tractable microstate; richer tasks would introduce additional slow and hidden coordinates.

15 Discussion

The scientific value of this framework lies in compression with explicit boundaries. A local response coefficient connects delivered state to a network coupling. Magnetization identifies order, susceptibility identifies collective fluctuation, autocorrelation identifies persistence, and multi-information separates dependence from local uncertainty. Effective energy combines spatial, cross-layer, and field mismatch. Path reversal detects time asymmetry that may be invisible to static state frequencies. The strong field-reversal trajectory shows how these coordinates track reorganization and recovery. The overlap of partition, corruption, and nominal trajectories shows the complementary boundary: a fixed reduced representation does not guarantee sensitivity to a weak or dynamically ineffective perturbation.

The immutable V12 nonreciprocity boundary is also informative. Directed delivery alone was not the dominant detectable source of time asymmetry under matched degree and traffic. In contrast, bounded memory explicitly introduces a historical coordinate and is a plausible mechanism for persistence and irreversibility. This motivates future theory in which message semantics and memory are slow variables rather than treating all asymmetry as an antisymmetric pair coupling.

Future work should add independent models, richer finite-state decisions, larger direct sizes, longer stationary records, and preregistered semantic perturbations. Exact small-state kernels could clarify which projected currents are robust to hidden variables. None of these extensions is needed to interpret the present finite-size measurements, but they are needed before universality or thermodynamic-limit claims.

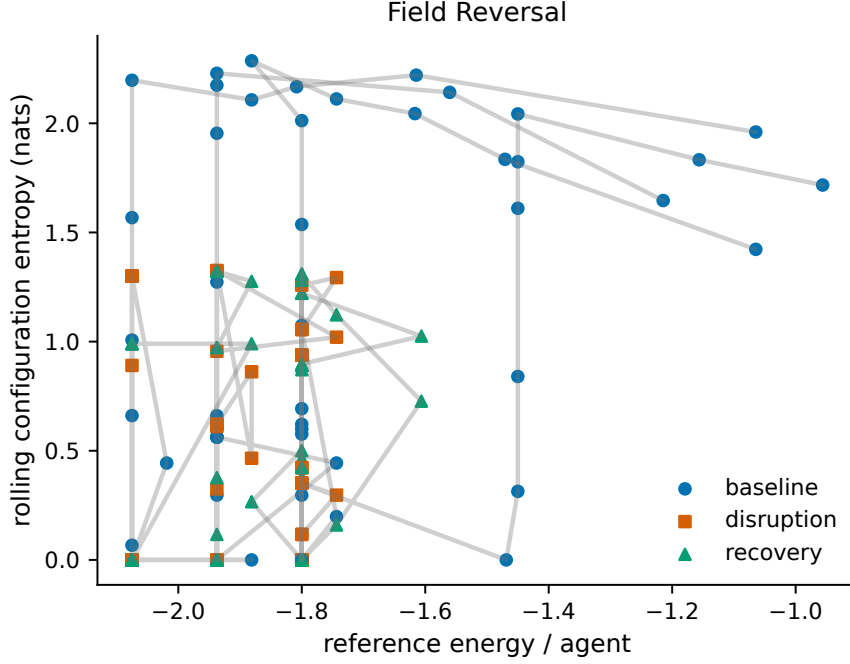


Figure 9: Energy–entropy paths during field reversal. Color and marker distinguish baseline, disruption, and recovery; lines retain temporal ordering.

16 Conclusion

Independent LLM agents with private state and explicit message delivery form a measurable stochastic many-agent process. The V12 coupling and decoding-noise directions did not replicate: V13 estimates were null or reversed, and the fitted kinetic surrogate captured only the noise direction. Bounded memory did replicate, increasing adjusted pathwise irreversibility by 0.04030 nats per attempted update. The reduced energy–entropy–order representation sharply tracked field reversal and improved four-class accuracy over simpler representations, but separation of partition, corruption, and nominal operation remained weak. Statistical mechanics is therefore useful here as a finite-size, reduced descriptive framework with demonstrated memory and quench sensitivity, not as a universal response law or a promise that thermodynamic quantities improve agent performance.

Reproducibility and data availability

The protocol, aggregate tables, figure source data, vector figures, source, tests, checksums, and exact commands accompany the repository. Raw prompts, completions, and attempted-update records are excluded from Git and retained in a checksummed external artifact tree. Formal execution used 14.74 metered generation GPU-hours. Deterministic replay feeds each content-addressed model decision through the same transition code without another model call. The repository reports model revision, quantization, package versions, tokens, calls, byte counts, exclusions, and every independent graph cluster.

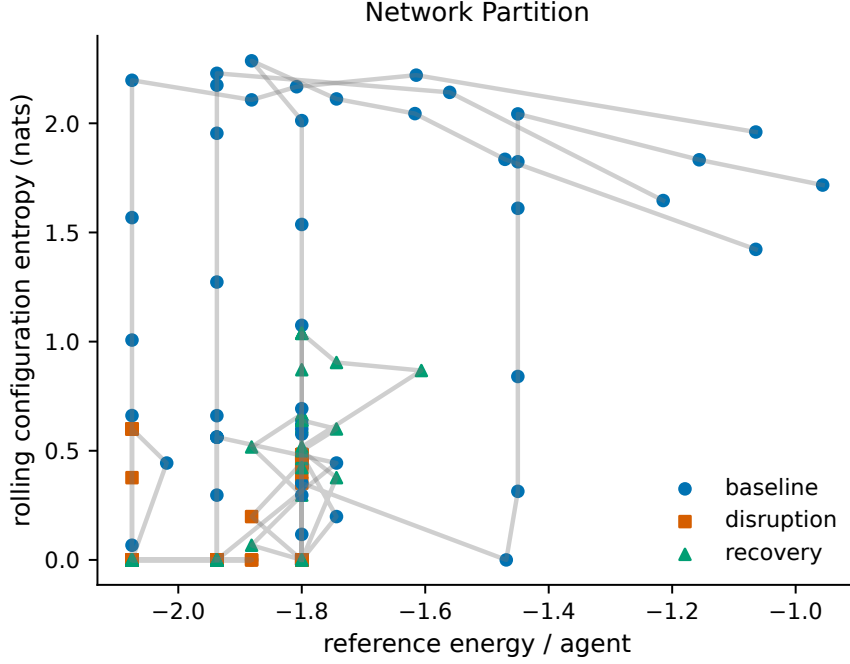


Figure 10: Energy-entropy paths during removal and restoration of inter-community delivery links. The reference-energy layer itself remains fixed.

A Random-sequential transition protocol

For every sweep, a deterministic seed generates a random permutation of all N identities. At opportunity k , only the scheduled agent’s prompt is constructed. Its inbox contains the latest packets actually delivered to that identity; a peer’s private field, memory, and undelivered outbox are absent. The validated model object is decoded through the panel’s latent-label map and applied to that agent. Its selected packet is serialized, one recipient is drawn from the current delivery row, and delivery occurs before opportunity $k + 1$. Consequently, updates within one sweep are temporally linked, while each identity receives exactly one opportunity per sweep.

Matched arms reuse graph, initial state, private fields, label map, opportunity permutations, recipient variates, and model seeds. The perturbation changes only the declared factor. During a partition, cross-community paths are removed and remaining row weights renormalized. During corruption, categorical packet fields from the preassigned even-index sender half are flipped after the agent constructs the packet; the agent-owned decision and outbox are not rewritten. Restoration resets fields and delivery exactly and disables the channel transform.

B Information estimators

Let Y_t denote the fixed coarse macrostate word. For observed counts $n(y_{t-k:t})$, the order- k conditional entropy estimator is

$$\hat{h}_k = - \sum_{y_{t-k:t-1}} \hat{p}(y_{t-k:t-1}) \sum_{y_t} \hat{p}_\delta(y_t | y_{t-k:t-1}) \log \hat{p}_\delta(y_t | y_{t-k:t-1}), \quad (14)$$

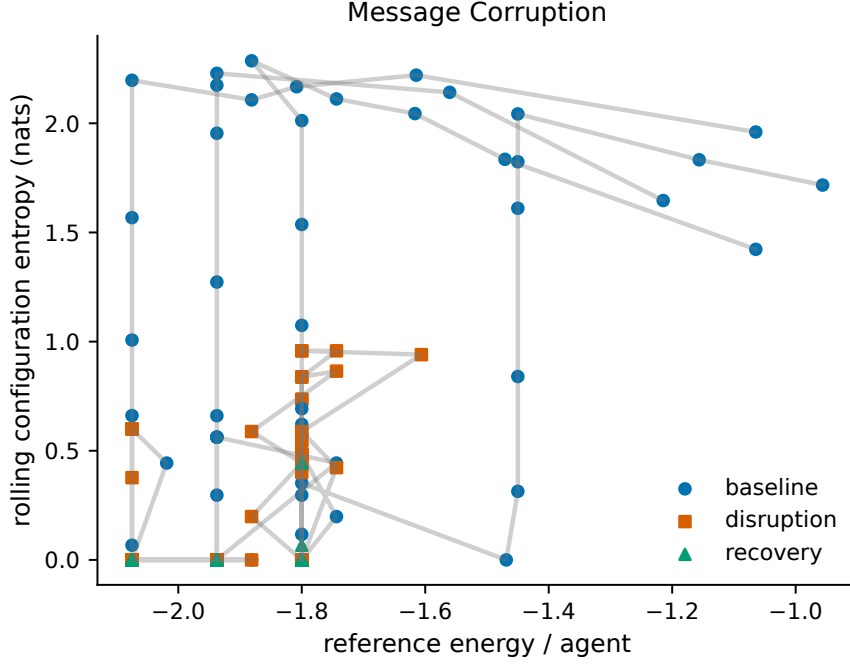


Figure 11: Energy–entropy paths under matched-volume categorical message corruption and subsequent clean-channel recovery.

where δ is the frozen pseudocount. Total correlation is evaluated on the joint belief–action word over the same rolling window. These plug-in quantities describe the chosen coarse-graining; neither is extrapolated to an unobserved microscopic entropy.

The reversal estimator forms length- L words, computes smoothed forward and reverse empirical probabilities on their union support, and divides the KL divergence by $L - 1$ attempted-update intervals. Two hundred within-trajectory time permutations establish the finite-length occupancy-matched floor. The primary adjusted statistic subtracts that mean without truncation, so negative values are possible. Confidence intervals resample complete matched clusters, not overlapping words.

C Trajectory-level representation comparison

The disruption classifier receives one feature vector for each complete panel, constructed from baseline-to-disruption changes and recovery residuals. All four panels from one graph/environment cluster are withheld together. Scaling and multinomial-logistic coefficients are fit only on the other clusters. Thus windows from a held-out trajectory never enter model fitting, nominal-manifold estimation, or feature selection. The simple, order-only, and full feature sets are frozen in the protocol; their comparison is an ablation of collective representation, not a controller competition.

D Agent autonomy and information-boundary audit

Agent autonomy is an experimental constraint rather than a label attached to the simulator. Each identity is represented by a separate state object and a separate bounded inbox, outbox, memory list, private field, workload, belief, action, confidence, and commitment.

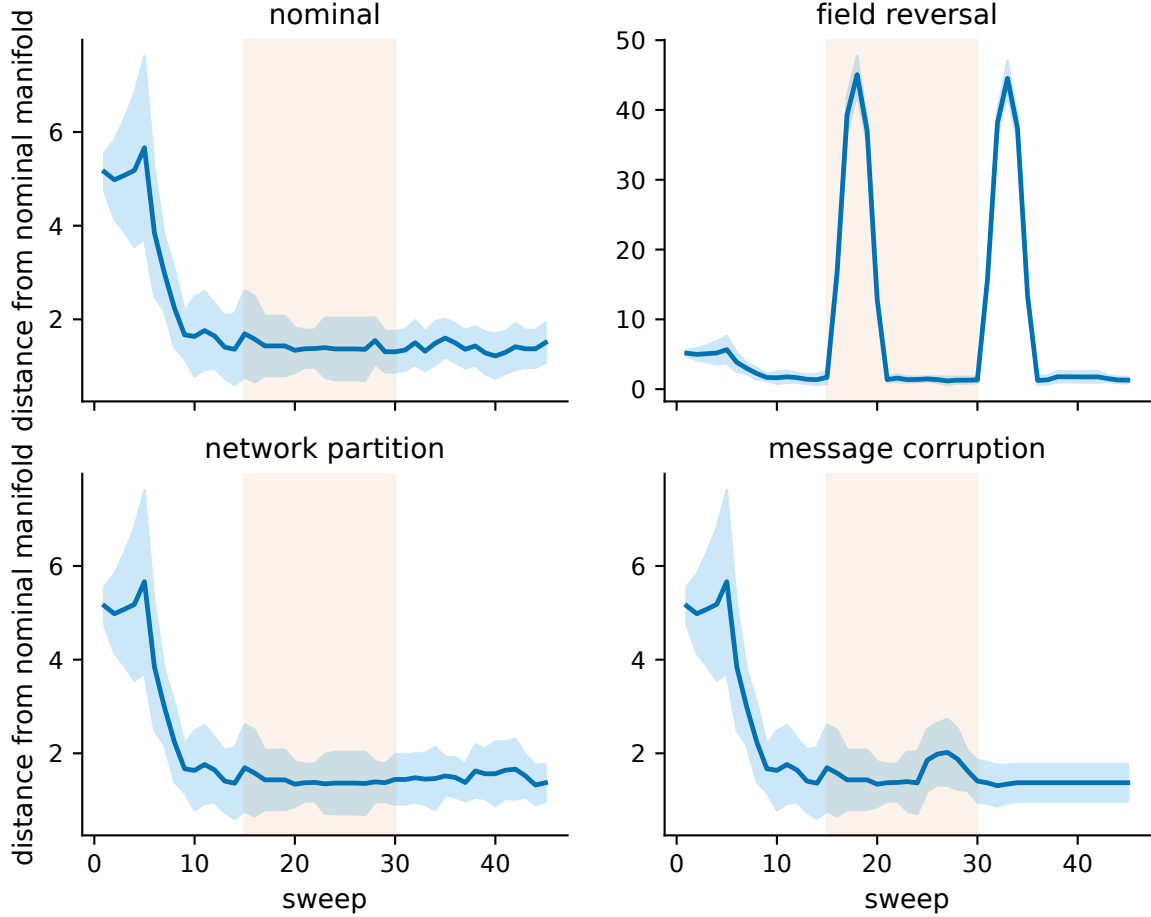


Figure 12: Distance from the nominal reduced-state manifold across the four matched conditions. Shading shows the graph-cluster standard error; the colored span is the prospectively fixed disruption interval.

Prompt construction accepts the scheduled identity and a read-only view of that object’s delivered information. It has no argument through which evaluator-only X_t , peer memory, or an undelivered packet can enter. The scheduler supplies only an update opportunity and a recipient draw. It does not supply a proposed scientific state.

Three counterfactual tests enforce this boundary. First, perturbing one agent’s private field or memory may change that agent’s serialized prompt but must leave every peer-private digest invariant. Second, a deliberately invalid model object may trigger one syntax repair, after which failure produces a recorded no-change transition rather than a central replacement. Third, partition and restoration modify only the delivery operator: agent-owned outboxes and state remain unchanged. Formal trajectories record peer-mutation counts, message opportunities, delivered packets, complete wire bytes, and content-addressed raw-decision digests. Any privacy mutation or scheduler substitution invalidates the run rather than becoming an exclusion.

Latent alternatives are counterbalanced independently of the scientific factors. Display order, the mapping from displayed words to ± 1 , initial beliefs and actions, and private-field signs are balanced by frozen seeds. The model output is decoded only after schema validation. Consequently, a display-side preference remains measurable as a nuisance response but cannot systematically define high coupling, high noise, memory, or disruption arms.

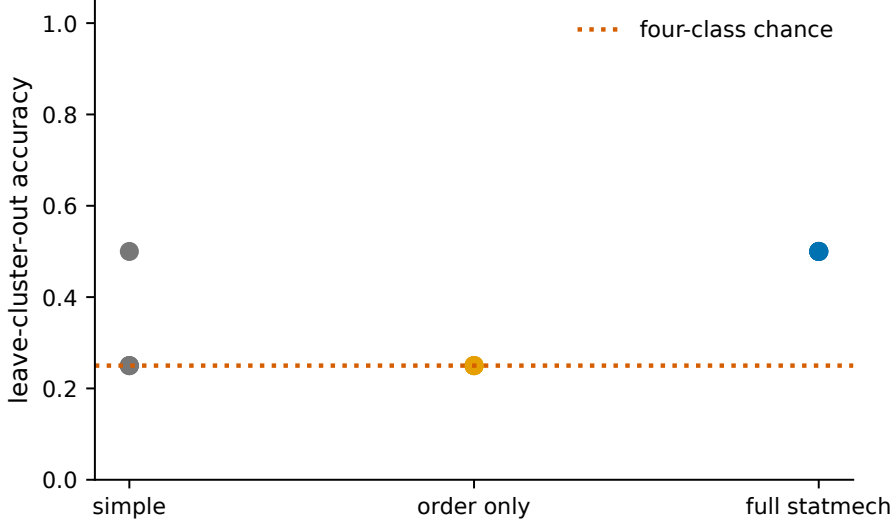


Figure 13: Leave-one-graph-cluster-out disruption classification for simple, order-only, and full statistical-mechanical representations. Points are held-out clusters, and the dotted line is four-class chance.

E Quench operators and matched restoration

Let $\mathcal{P}_0$ denote the baseline transition operator induced by the fixed graph, local fields, update schedule, and clean message transform. Each quench replaces exactly one component during sweeps 16–30 and restores the original object for sweeps 31–45. Field reversal uses $\mathbf{h} \mapsto -\mathbf{h}$ while leaving communication unchanged. The partition uses a masked delivery matrix

$$W_{ij}^{\text{part}} = W_{ij} \mathbf{1}[c(i) = c(j)] \quad (15)$$

followed by row normalization over remaining eligible neighbors. Opportunity counts remain fixed even when a cross-community delivery is unavailable. Message corruption applies a deterministic content transform to the preassigned half of senders after packet serialization and before delivery; the number and encoded length of packets are unchanged. The undisturbed arm retains $\mathcal{P}_0$ throughout.

Recovery is therefore a return of the exogenous operator, not a reset of the agent configuration. Residual displacement measures state retained by the collective after the drive is removed. Because these are abrupt, known interventions, departure at or after the boundary is a response measure and cannot be interpreted as prospective early warning.

F Frozen hypothesis tests and protocol amendments

For H1, the cluster-level contrast is the mean of all matched $J = 0.80$ minus $J = 0.35$ trajectories within a modular graph/environment cluster, separately for absolute order, susceptibility, and correlation time. H1 is supported only when all three directions are positive by interval and the maximum one-sided endpoint p value survives familywise correction. H2 uses the analogous $\tau = 0.85$ minus $\tau = 0.50$ contrast and requires all three directions to be negative. H3 uses one persistent-minus-Markovized contrast per matched cluster. Exact sign-flip randomization and 10,000 cluster-bootstrap resamples preserve the paired hierarchy. Holm correction is applied to the H1–H3 hypothesis-level values; secondary endpoints are not used to rescue a failed primary claim.

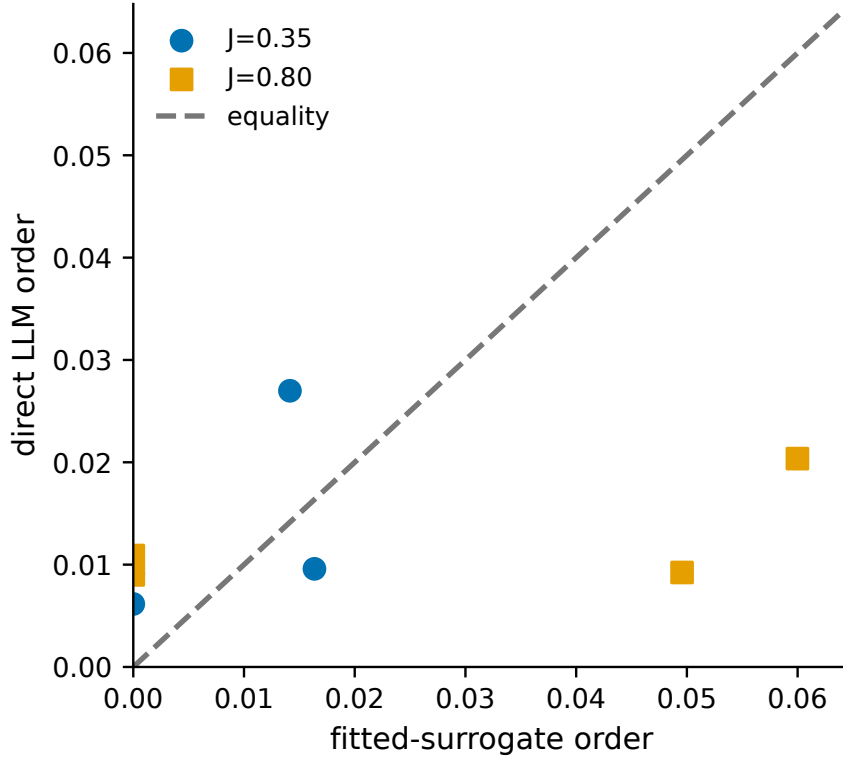


Figure 14: Direct LLM order against the fitted decentralized kinetic surrogate at matched anchors. The diagonal is equality, not a fit target.

Two administrative amendments preceded the operative freeze and are retained verbatim in the package. During the incomplete microscopic block under version 1.0, it was noticed that four paired H3 clusters imply a minimum exact one-sided sign-flip probability $1/16 = 0.0625$, making the declared 0.05 rule mathematically unattainable. No graph trajectory or contrast then existed. Generation stopped after 163 microscopic records; all were retained and the H3 count was changed prospectively to six while secondary computation was reduced. A second pre-restart amendment made the global prompt-token cap explicitly include pilot and interrupted records. Neither amendment altered an observed network outcome, estimator, effect direction, or application of a criterion.

G Nominal manifold and trajectory geometry

For held-out quench cluster g , the nominal center $\hat{\mu}_{-g}$ and ridge-regularized covariance $\hat{\Sigma}_{-g}$ are estimated only from undisturbed trajectories in the other clusters. The standardized departure is

$$d_{g,t} = \left[(Z_{g,t} - \hat{\mu}_{-g})^T \hat{\Sigma}_{-g}^{-1} (Z_{g,t} - \hat{\mu}_{-g}) \right]^{1/2}. \quad (16)$$

Its nominal 95th percentile is likewise learned without cluster g . Maximum departure, time outside that region, final residual, and the first two consecutive recovery points below threshold are computed at trajectory level. The energy–entropy loop area uses the polygonal path in temporal order and is a finite dynamical hysteresis descriptor; it is not thermodynamic work.

The representation comparison uses baseline-to-disruption differences and recovery

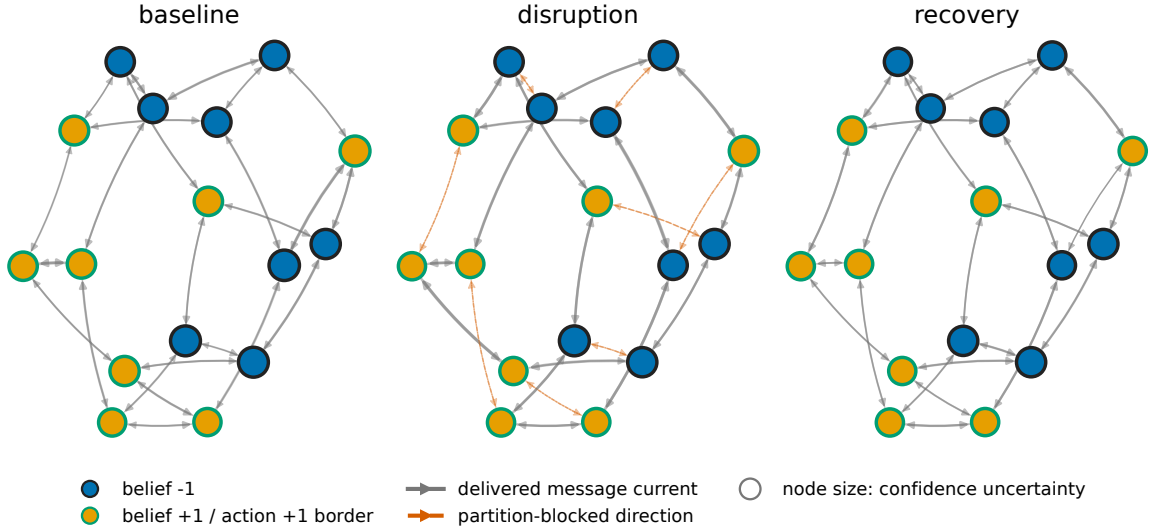


Figure 15: Agent beliefs, actions, uncertainty, and realized communication before, during, and after the fixed inter-community partition. Node fill denotes belief, border denotes action, size denotes confidence uncertainty, and edge width denotes message current.

residuals. All four conditions sharing one graph and environment are withheld together. A fixed L_2 -regularized multinomial logistic model and training-fold standardization prevent the full representation from receiving a different model family. The analysis reports accuracy, log loss, graph-level predictions, and the paired full-minus-strongest-reduced contrast. With four held-out clusters this is a deliberately modest descriptive test, so coefficient patterns are not promoted to universal biomarkers.

H Estimator and coarse-graining sensitivity

The primary block length is three and pseudocount is 0.5. Sensitivities use block lengths two and four and pseudocounts 0.1 and 1.0. Configuration bins are fixed before formal execution. Time-shuffled paths preserve length and marginal occupancy while destroying temporal order; their mean reversal divergence is subtracted as the primary bias floor. These corrections can yield a slightly negative adjusted value, which is retained rather than truncated.

I Additional candidate figures

The paper package contains 22 candidate vector figures with exact source CSVs. Supplemental candidates include nominal portraits, magnetization–entropy paths, network snapshots, finite-size comparisons, memory-loop areas, discovery–confirmation forests, and an observable–redundancy map. The figure catalog records each plot’s scientific purpose, source table, suggested role, supported claim, and limitation. Selection for submission should follow scientific review rather than visual favorability.

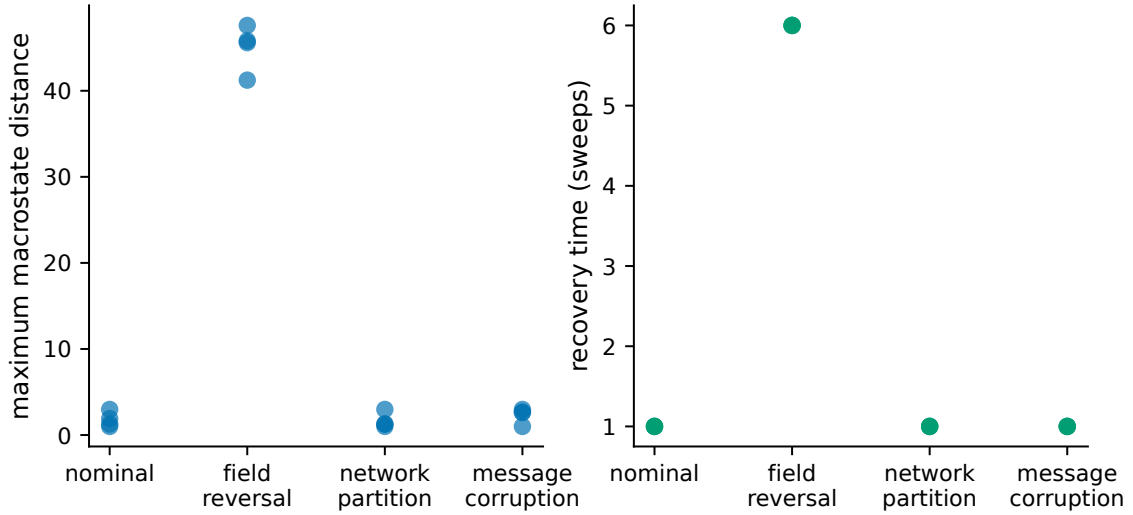


Figure 16: Graph-level maximum departure and recovery time under the four matched operators. Distances use a leave-cluster-out nominal reference and a frozen regularization rule.

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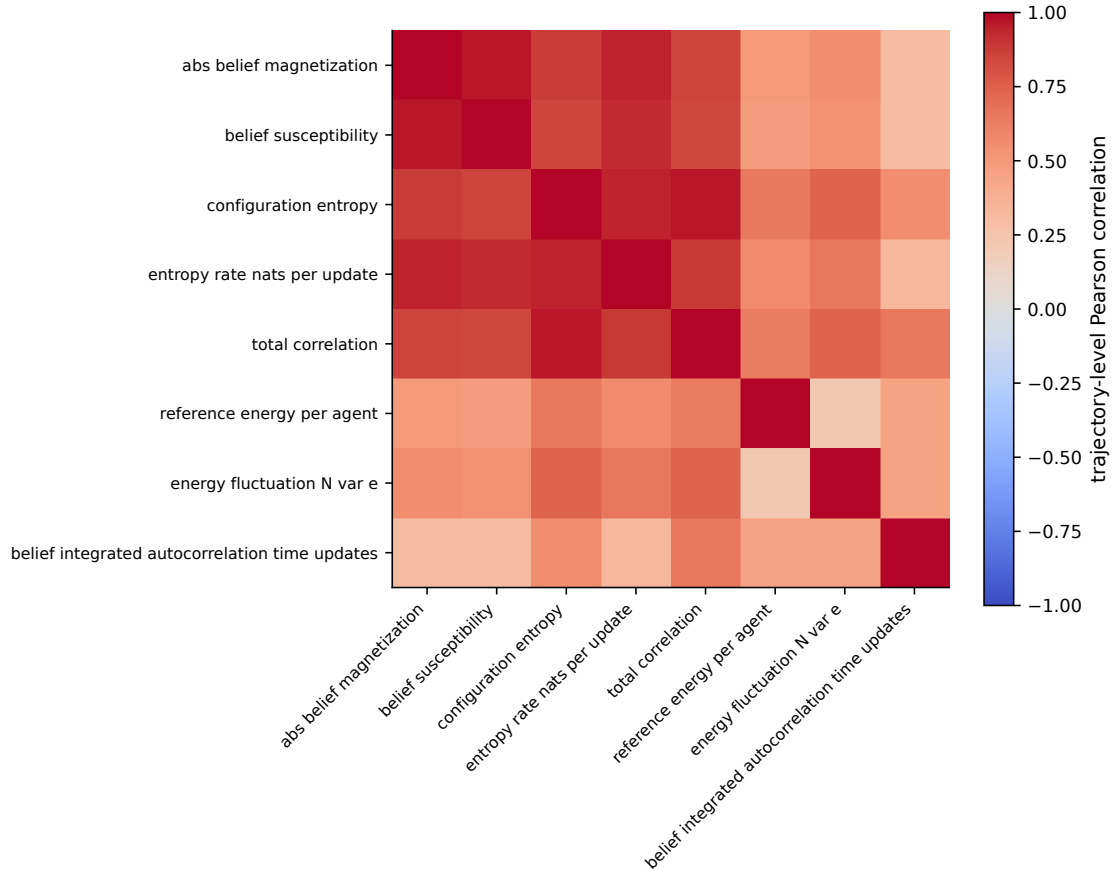


Figure 17: Trajectory-level redundancy among order, fluctuation, information, effective energy, and persistence coordinates. Correlation does not itself determine which variable is causally upstream.

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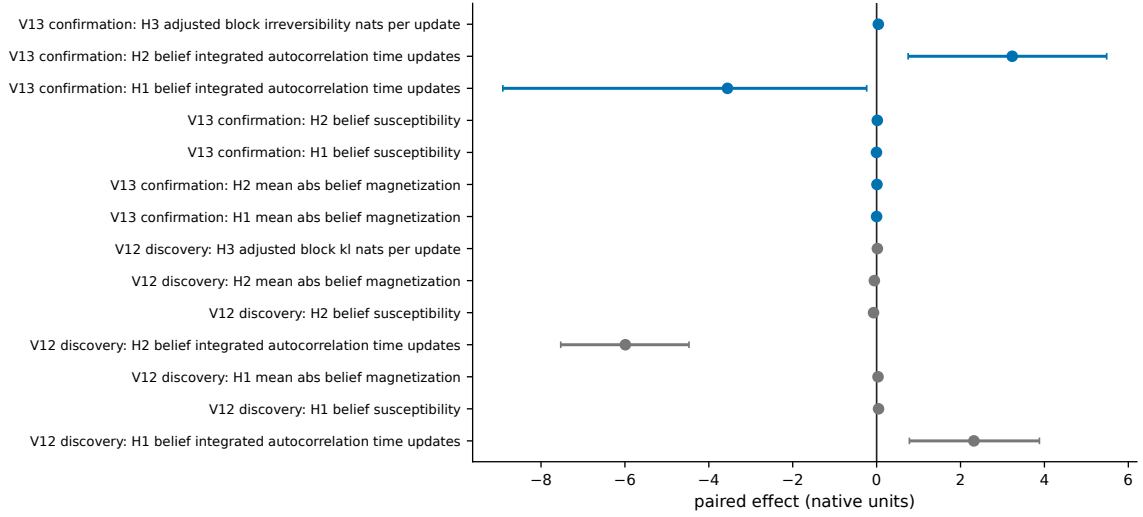


Figure 18: Immutable V12 discovery estimates and independent V13 confirmation estimates. Native units differ across endpoints; intervals resample graph-level clusters.

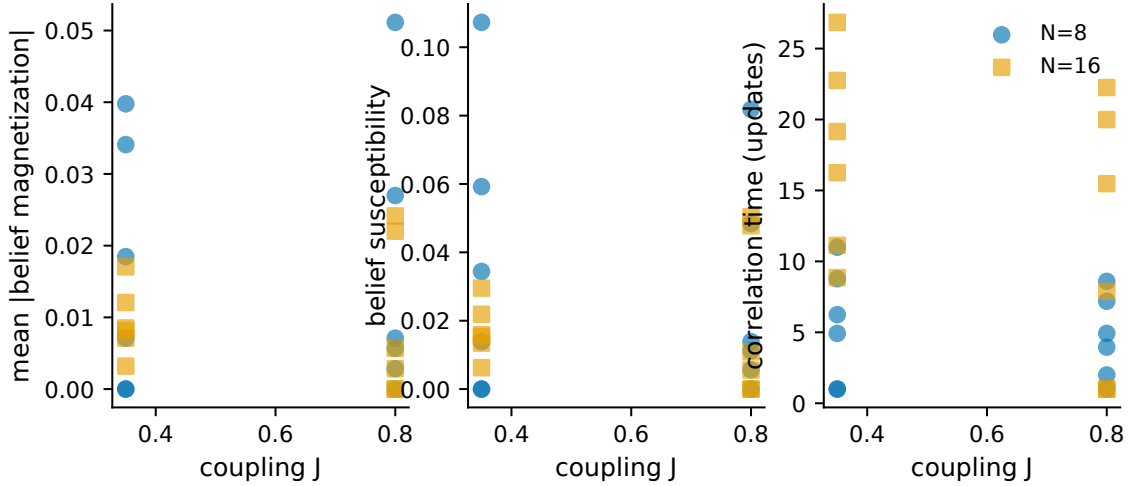


Figure 19: Direct $N = 8$ and $N = 16$ order, susceptibility, and correlation-time diagnostics. The comparison is finite-size evidence, not scaling collapse.

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